

The empirical recurrence for OEIS A301612, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A301612 records “Empirical recurrence of order 68” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 70$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 70$. Its last coefficient is $c_{68} = -8100 \neq 0$, so no further tail satisfies anything shorter; any order-68 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A301612 is “Number of $n \times 5$ 0..1 arrays with every element equal to 1, 2 or 3 horizontally or vertically adjacent elements, with upper left element zero.”. It has offset 1 and begins

3, 111, 2131, 40085, 758338, 14345706, 271301458, 5131197358, ...

The entry states:

Empirical recurrence of order 68 (see link above)

As of the “Last modified” line on the live entry (Mar 24 2018 11:40:34, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 70$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 70$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 68: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 68.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 140$ exact terms from $n = 2$ on, it returns order 68 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{68} c_i a(n-i),$$

c_1	11	c_{18}	−89404331539	c_{35}	17537104665013	c_{52}	25487417229
c_2	180	c_{19}	412942371817	c_{36}	6022485335549	c_{53}	−3529820329
c_3	−175	c_{20}	1092498788317	c_{37}	2171941867034	c_{54}	7255204429
c_4	−7574	c_{21}	−981704695083	c_{38}	511172658526	c_{55}	356000656
c_5	−8959	c_{22}	−6415125426091	c_{39}	−5609627315121	c_{56}	−1757974486
c_6	164664	c_{23}	−1715696390822	c_{40}	2387000022538	c_{57}	−959545702
c_7	421507	c_{24}	20960762063507	c_{41}	2372434415449	c_{58}	143115830
c_8	−2066231	c_{25}	7123060200022	c_{42}	−2254905711522	c_{59}	78626888
c_9	−8136421	c_{26}	−56194368444216	c_{43}	305408420688	c_{60}	135769406
c_{10}	16396856	c_{27}	−6035051383643	c_{44}	475620870112	c_{61}	−22398550
c_{11}	99414684	c_{28}	90335534101641	c_{45}	−687949216726	c_{62}	−15699790
c_{12}	−90654402	c_{29}	−7339598541159	c_{46}	177703787881	c_{63}	−3893240
c_{13}	−988768832	c_{30}	−87208217276955	c_{47}	133874018591	c_{64}	484855
c_{14}	175966367	c_{31}	14373210454483	c_{48}	140197924118	c_{65}	636900
c_{15}	8632954316	c_{32}	77754415561998	c_{49}	−20337790253	c_{66}	195525
c_{16}	3729368795	c_{33}	−22701772498577	c_{50}	−79247193835	c_{67}	−86850
c_{17}	−68899196416	c_{34}	−46918267018689	c_{51}	−19429424333	c_{68}	−8100

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 70$, and it is the recurrence OEIS A301612 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{68} - c_1 x^{67} - \dots - c_{68}$. Its constant term is $-c_{68} = 8100$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 68 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 68, so $\deg q = 68$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 68, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 68, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -8100 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-68 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A301612.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.